

Speed, Distance & Time



What Is Speed?

Speed tells us how FAST something moves.

If you run quickly, your speed is high.

If you walk slowly, your speed is low.

Speed is measured in km/h or m/s



What Is Distance?

Distance is how FAR you travel from one place to another.

It could be from your home to school, or from one city to another!

Distance is measured in meters (m) or kilometers (km)



What Is Time?

Time tells us how LONG it takes to go from one place to another.

We measure time in seconds, minutes, and hours.

Time is measured in seconds (s), minutes (min), hours (h)



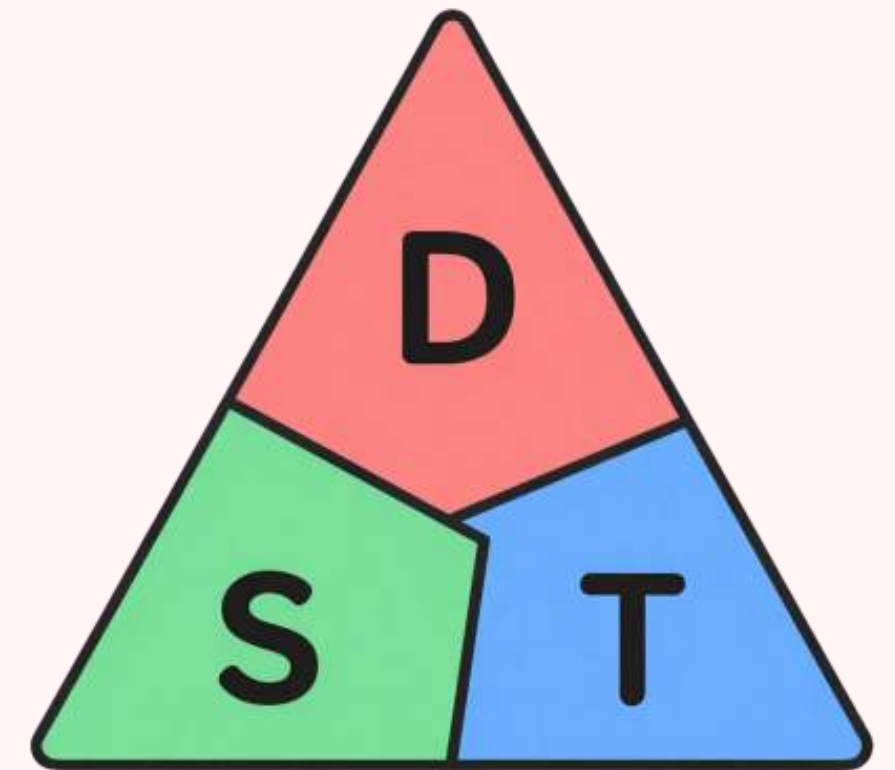
The Magic Formula!

$$\text{Speed} = \text{Distance} / \text{Time}$$

$$\text{Distance} = \text{Speed} \times \text{Time}$$

$$\text{Time} = \text{Distance} / \text{Speed}$$

Remember: $S = D / T$
Just think of it as a triangle!



Example 1: Walking to School

Given:

Distance = 2 km

Time = 30 minutes

Solution:

Speed = Distance / Time

Speed = $2 / 0.5 = 4$ km/h



Example 2: Cycling in the Park

Given:

Distance = 10 km

Time = 20 minutes

Solution:

Speed = Distance / Time

Speed = $10 / (20/60) = 30 \text{ km/h}$



Example 3: Running Race

Given:

Distance = 100 meters

Time = 20 seconds

Solution:

Speed = Distance / Time

Speed = $100 / 20 = 5 \text{ m/s}$



Quiz Time! Question 1

Riya walks 4 km in 2 hours.

What is her speed?

- A) 4 km/h
- B) 2 km/h
- C) 8 km/h

Answer: B) 2 km/h ($4 / 2 = 2$)



Quiz Time! Question 2

A bus travels at 60 km/h for 3 hours.

How far does it go?

- A) 20 km
- B) 180 km
- C) 63 km

Answer: B) 180 km ($60 \times 3 = 180$)



Quiz Time! Question 3

Arjun cycles 15 km at 5 km/h.

How long does it take?

- A) 75 hours
- B) 10 hours
- C) 3 hours

Answer: C) 3 hours ($15 / 5 = 3$)



Real Life: School Bus

Your school bus travels 12 km in 20 minutes every morning.

The driver uses speed, distance & time to make sure you reach school on time!

Speed helps plan journeys!



Real Life: Train Journey

The Delhi to Jaipur train covers
280 km in about 4 hours.

Speed = $280 / 4 = 70$ km/h

Trains use speed to keep schedules!

Faster trains save travel time!



What Did We Learn?

1. Speed tells us how FAST we move
2. Distance tells us how FAR we travel
3. Time tells us how LONG it takes
4. Speed = Distance / Time
5. We can solve real-life problems with this!

You are now a Speed, Distance & Time expert!



Great Job!

You learned Speed, Distance & Time today!
Do you have any questions?

Keep exploring and stay curious!



Bonus: Speed Challenge!

Try this fun activity at home!

1. Mark 10 metres on the ground
2. Run from start to finish
3. Ask a friend to time you with a stopwatch
4. Calculate: Speed = $10 \text{ m} / \text{your time}$

Share your speed with your classmates!

